Professor Q&A Notes: Taxonomy-aware BiomeGPT, Prediction Results, and Batch Correction

Yuanshu / Codex working report

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How To Use This Document

This is not the main technical report. It is a meeting-preparation document: likely questions, direct answers, and the reasoning behind our choices. The main message is that we have three connected pieces of work:

1. A taxonomy-aware BiomeGPT model for microbiome representation learning.
2. A Healthy-vs-Diseased prediction pipeline with external validation.
3. A batch-effect diagnostic and preliminary adversarial correction pipeline.

1 One-Minute Summary

What did we build? We reproduced a BiomeGPT-style microbiome transformer and extended it with taxonomy-aware species embeddings. Each sample is represented by a sample-level `<cls>` embedding, and each microbial species has a learned species embedding informed by taxonomy.

Why taxonomy-aware? Microbial species are not independent IDs. Species from the same genus, family, order, or phylum share biology. Adding rank-wise taxonomy embeddings gives the model an inductive bias that should improve interpretability and transfer when external cohorts have sparse or shifted species coverage.

Why batch diagnostics? External validation can fail because the model learns study/cohort signals instead of robust disease biology. We reconstructed study/cohort batch-proxy labels from public accession metadata and tested whether the pretrained embeddings encode those labels.

What did we find? Batch signal is present, but broad batch labels are strongly phenotype-confounded. Naive adversarial correction using broad labels can fail or even increase post-hoc batch separability. Conservative-safe labels remain the best correction target, but adversarial optimization needs careful scheduling.

2 What Is The Biological Task?

Likely question: What biological problem are we solving?

We are studying whether gut microbiome species-abundance profiles can be represented by a foundation-model style transformer and used for Healthy-vs-Diseased prediction. A gut microbiome sample is a sparse vector of microbial species abundances. The biological challenge is that disease-associated microbial patterns are subtle, cohort-dependent, and strongly affected by geography, sequencing protocol, study design, and phenotype mixture.

Answer to give:

> We are not only training a binary classifier. We are trying to learn a transferable microbiome representation: species-level embeddings and sample-level embeddings that capture community structure, taxonomy, and disease-associated shifts across cohorts.

3 What Is BioGPT / BiomeGPT Doing?

Likely question: How does the model represent a microbiome sample?

In our implementation, each sample is a sequence of species tokens. Each token has two parts:

- a species identity representation;
- an abundance-bin representation.

The model uses masked abundance reconstruction during pretraining. We randomly mask a fraction of observed nonzero species abundances and ask the transformer to reconstruct their abundance bins from the rest of the microbial community. This is analogous to masked language modeling, except the tokens are microbial species and the prediction target is an abundance bin rather than a word.

Key output representations:

- **Species prompt**: learned species representation, analogous to a gene embedding in scGPT.
- **Sample prompt**: final `<cls>` embedding for a microbiome sample, analogous to a cell embedding in scGPT.

4 What Did We Change: Taxonomy-aware BiomeGPT

Likely question: What is new compared with a plain species-ID model?

The original species token can treat every species as an unrelated ID. We changed the species representation to include taxonomy:

```
Domain + Kingdom + Phylum + Class + Order + Family + Genus + Species
```

This means two species from the same genus share the genus component; species from the same family share the family component; and so on.

Why this is biologically reasonable:

- Microbial traits are partially structured by taxonomy.
- External datasets may contain related but not identical species.
- Taxonomy-aware embeddings make the species latent space more interpretable.
- The inductive bias may help transfer when species overlap across cohorts is imperfect.

5 What Data Did We Use?

Data component	Count / role
Phase-1 pretraining species vocabulary	1,012 species
Phase-2 gut samples for pretraining / batch diagnostics	13,332 samples
Fine-tuning training set	_prev3 gut data
External validation set	927 samples
ExVal label balance	612 Healthy / 315 Diseased
Conservative-safe batch-correction samples	2,134 samples

The ExVal set overlaps with 512 of the 513 `_prev3` fine-tuning species, so external evaluation is feasible after reindexing species columns to the training vocabulary.

6 Pipeline Overview

1. **Phase 1 pretraining**: gut + non-gut samples, learn broad microbiome ecological structure.
2. **Phase 2 pretraining**: gut-only adaptation, make the representation more domain-specific for stool/gut samples.
3. **Embedding extraction**: extract species prompts and sample prompts.
4. **Healthy-vs-Diseased fine-tuning**: add classifier on sample prompt and train on `_prev3`.
5. **External validation**: evaluate on ExVal using accuracy, F1, AUROC, macro-accuracy, macro-F1, and class-specific H/D accuracy.
6. **Batch diagnostics**: test whether phase-2 sample prompts encode study/cohort labels.
7. **Adversarial correction**: preliminary GRL-based correction on conservative-safe batch labels.

7 Important Parameters And What They Mean

Parameter	Meaning
`bins = 32`	Continuous relative abundances are converted into 32 rank-based abundance bins. This reduces scale sensitivity across datasets and turns abundance prediction into a structured masked reconstruction problem.
`mask_ratio = 0.25`	During pretraining, 25% of nonzero observed species are masked and reconstructed. This forces the model to learn community context rather than memorizing each input value.
`d_model = 512`	Hidden embedding dimension for species/sample representations. Larger values can capture more structure but increase training cost and overfitting risk.
`nhead = 8`	Number of transformer attention heads. Multiple heads allow the model to learn different interaction patterns among species.

Parameter	Meaning
`num_layers = 8`	Number of transformer blocks. More layers increase capacity but also training instability and overfitting risk.
`lr`	Learning rate. Pretraining used a larger learning rate than fine-tuning/correction because fine-tuning starts from an already learned checkpoint.
`threshold`	Probability cutoff for calling a sample Diseased in H/D prediction. This strongly affects Healthy accuracy vs Diseased accuracy.
`macro-F1`	Average of class-wise F1. Important because Healthy and Diseased counts are imbalanced.
`AUROC`	Threshold-independent ranking metric. It can be good even if a chosen threshold gives poor Healthy/Diseased balance.
`DAB weight`	Weight on the adversarial batch-discriminator loss. Larger is not automatically better.
`GRL lambda`	Strength of gradient reversal. It controls how strongly the encoder is pushed to hide batch information.
`min_batch_size = 10`	Minimum samples required for a batch label to be used in a probe/correction trial. This avoids unstable tiny classes.

8 Why Did We Need Batch Labels?

Likely question: Did the original metadata contain batch labels?

No. The original phase-2 gut metadata did not contain explicit sequencing batch, study ID, cohort ID, sequencing center, platform, country, library prep, or run-date columns. It contained phenotype and sample IDs. Therefore, we reconstructed a study/cohort-level batch-proxy annotation.

How we reconstructed labels:

- Public accession IDs such as `SAMEA`, `SAMN`, `SAMD`, `SRR`, and `EGAR` were queried against ENA, NCBI SRA/BioSample, and EGA.
- Local IDs without public accessions were assigned prefix-derived labels only when the pattern was structured and interpretable.
- Every label received confidence: high, medium, or low.
- Every batch label was checked for phenotype confounding.

Confidence	Samples	Interpretation
High	9,389	Exact public accession lookup returned study/cohort metadata
Medium	3,418	Structured local prefix-derived cohort label
Low	525	Unresolved or ambiguous local sample ID pattern

9 What Is Batch-Phenotype Confounding?

Likely question: If the batch label is real, why not correct it directly?

A real study label is not automatically safe. Many microbiome studies are phenotype specific. For example, one study may contain only Healthy samples, while another study may contain mostly Parkinson's disease or colorectal cancer samples. If we remove study signal naively, we may also remove true disease biology.

Rule used:

> `phenotype_confounding_warning=True` if the top phenotype fraction in a batch is `>= 0.80` or if the batch has only one phenotype.

Observed confounding:

Label panel	Cramer's V	NMI	Median top phenotype frac.	Batches top frac. ≥ 0.8
High + medium	0.802	0.519	1.000	79 / 105
High-only	0.907	0.489	1.000	66 / 83
Conservative-safe	0.683	0.457	0.549	0 / 17

Answer to give:

> We can use broad labels for diagnostics, but final correction should use only labels that are high-confidence and not strongly phenotype-confounded.

10 What Is A Batch Probe?

Likely question: What does the batch probe actually measure?

A batch probe is a shallow classifier trained on frozen sample embeddings. The model itself is not changed. If a logistic regression classifier can predict batch labels from the `<cls>` embedding, then the embedding contains batch-associated information.

Probe	Samples	Classes	Balanced acc.	Macro-F1
High + medium batch	12,758	105	0.397	0.365
High-only batch	9,340	83	0.386	0.355
Conservative-safe batch	2,134	17	0.503	0.499
Healthy vs Diseased probe	13,332	2	0.801	0.796

Important distinction: the H/D probe is not the final ExVal classifier. It is a diagnostic test showing that frozen pretrained embeddings contain H/D-separable signal in the phase-2 data.

11 How Did We Implement Batch Correction?

We adapted scGPT-style domain-adversarial training:

- Attach a small batch discriminator to the `<cls>` sample embedding.
- Use a Gradient Reversal Layer (GRL).
- Keep the original masked abundance reconstruction loss.
- Add a batch-discriminator loss with small weight.

The light correction objective was:

```
total loss = reconstruction loss + 0.1 * batch cross-entropy loss
```

Because the batch branch uses GRL, the discriminator learns to predict batch, while the encoder receives reversed gradients and is pushed to make `<cls>` less batch-predictive.

12 What Did Batch Correction Show?

Run	Batch F1 before	Batch F1 after	H/D F1 before	H/D F1 after
Conservative-safe, batch size 8, DAB 0.1, 3 epochs	0.499	0.476	0.796	0.789
High + medium, batch size 32, DAB 0.1, 3 epochs	0.365	0.383	0.796	0.804
High-only, batch size 32, DAB 0.1, 3 epochs	0.355	0.373	0.796	0.802
Conservative-safe, batch size 32, DAB 0.1, 3 epochs	0.499	0.511	0.796	0.794
Conservative-safe, DAB 0.5, 5 epochs	0.499	0.505	0.796	0.786

Interpretation: the first conservative-safe light run moved in the desired direction, but broader labels, larger batch size, and stronger DAB did not. This suggests the issue is not simply "train more." The adversarial game is sensitive to label quality, batch size, and schedule.

13 Improved Experiment: Evaluate The Pretraining Objective Directly

Likely question: Since BiomeGPT is a foundation model, should we evaluate batch correction only by H/D prediction?

No. H/D prediction is a downstream task. The core BiomeGPT pretraining task is masked abundance prediction. Therefore, batch correction should also be judged by whether the model preserves or improves masked abundance reconstruction, especially across study/cohort labels.

I added a reconstruction diagnostic:

```
dataset_v3/batch_reconstruction_diagnostics.py
```

It uses the same deterministic mask for each checkpoint, masks nonzero species positions, and evaluates mean squared error on the true abundance bins. It also computes per-batch MSE, batch MSE coefficient of variation, and batch MSE gap.

Checkpoint	Overall MSE	Safe-panel MSE	Safe-panel CV	Safe-panel gap
Original phase-2	34.857	37.808	0.095	13.140
Conservative-safe, batch size 8	33.745	35.724	0.093	12.961
Conservative-safe, batch size 32	34.088	36.252	0.096	13.324
High + medium, batch size 32	33.049	35.770	0.096	12.736
High-only, batch size 32	33.373	36.061	0.098	13.453
Conservative-safe, DAB 0.5	33.853	35.256	0.092	12.554

Interpretation: reconstruction MSE generally improved after continued correction training, so these trials did not destroy the masked-abundance objective. However, lower reconstruction MSE alone is not enough: some runs improved reconstruction while the batch probe got worse. Therefore, model selection should require three conditions:

1. batch probe decreases;
2. masked abundance reconstruction is preserved or improves;
3. H/D probe or downstream ExVal does not collapse.

Professor-facing wording:

> Because BiomeGPT is pretrained by masked abundance prediction, batch correction should not be evaluated only through disease classification. The primary foundation-model check is whether the corrected model remains good at masked abundance reconstruction, ideally with more balanced reconstruction error across batches. Disease prediction should be a downstream sanity check.

14 Why Was Prediction Around 0.5 Before, But Now Some Numbers Are 0.7–0.8?

This is the most important clarification. These numbers are not all the same metric.

Number	What it means	Comparable to ExVal macro-F1?
0.559	ExVal macro-F1 from selected full run at threshold 0.6	Yes
0.592	Fixed-threshold ExVal macro-F1 at threshold 0.8	Yes
0.609	Best value in threshold sensitivity diagnostic at threshold 0.9	Partly; diagnostic threshold sweep
0.769–0.783	ExVal AUROC	No; threshold-independent ranking metric
0.796	Frozen H/D probe macro-F1 on phase-2 embeddings	No; diagnostic probe, not ExVal prediction
0.96	Internal training-side calibration macro-F1 during selection	No; internal split, not external validation

Answer to give:

> The earlier 0.5–0.6 values are external validation macro-F1, which is the hardest and most relevant prediction metric. The 0.77–0.78 values are AUROC, not macro-F1. The 0.79 value is a frozen embedding H/D probe on phase-2 data, not the final external prediction. So these numbers are not contradictory; they measure different questions.

Why ExVal macro-F1 is lower than the internal calibration score:

- ExVal is a different external cohort with study/domain shift.
- The classifier tends to over-call Diseased at lower thresholds, causing low Healthy accuracy.
- Macro-F1 penalizes imbalance between Healthy and Diseased performance.
- Batch/study effects may make the learned disease signal less transferable.

15 How To Explain The ExVal Result

The selected run had:

Metric	Value
ExVal accuracy	0.563
ExVal F1	0.602
ExVal AUROC	0.783
ExVal macro-F1	0.559
Healthy accuracy	0.353
Diseased accuracy	0.971
Threshold	0.6

The fixed threshold 0.8 run improved Healthy specificity:

Metric	Value
ExVal accuracy	0.593
ExVal AUROC	0.770
ExVal macro-F1	0.592
Healthy accuracy	0.407
Diseased accuracy	0.956

Interpretation: the model ranks samples reasonably well by disease probability (AUROC around 0.77–0.78), but calibration/thresholding is poor under external shift. It predicts Diseased too often, which gives high Diseased accuracy but low Healthy accuracy. This is why macro-F1 is modest.

16 What If Professor Says Their Macro-F1 Is Around 0.7?

Answer carefully:

> That may be true, but we need to confirm the exact evaluation protocol: split, external cohort, threshold selection, class labels, species alignment, whether metrics are macro-F1 or AUROC, and whether threshold was tuned on external labels. Our current fully external macro-F1 is around 0.56–0.61 depending on threshold. We should not compare that directly to internal validation, AUROC, or an externally tuned threshold.

What to ask:

- Was the 0.7 value macro-F1, weighted F1, accuracy, or AUROC?
- Was it measured on ExVal or on an internal split?
- Was threshold selected on the training side only, or tuned using ExVal labels?
- Were Healthy/Diseased labels constructed in the same way?
- Were species columns aligned to the same vocabulary?
- Did they use the same phase-2 checkpoint and the same `_prev3` fine-tuning set?

17 What Should We Do If Batch And Phenotype Are Confounded?

Do not blindly remove all batch signal. If batch and phenotype are nearly the same variable, adversarial correction can erase disease biology.

Recommended approach:

1. Use broad labels only for diagnostics.
2. Use high-confidence, non-confounded labels for correction.
3. Keep post-correction H/D probes to check whether biological signal survives.
4. Use warm-up schedules for DAB/GRL instead of fixed full adversarial pressure.
5. Select correction checkpoints by a joint criterion: lower batch probe and preserved H/D probe.
6. Only evaluate final value by downstream ExVal H/D metrics.

18 Why Broad Batch Correction Failed

Our ablation suggests several possible mechanisms:

- Broad batch labels are heterogeneous and phenotype-confounded.
- The discriminator-encoder adversarial game may be unstable.
- The encoder may learn a new representation that is still batch-separable to a fresh post-hoc probe.
- There was no warm-up schedule; adversarial pressure was applied immediately.
- Batch size changed adversarial dynamics; batch size 8 and 32 behaved differently.

Meeting wording:

> The negative ablation is still informative. It shows that batch correction is not a one-line improvement. Label quality and phenotype confounding dominate. We need a cautious correction target and a scheduled adversarial optimization.

19 What Is The Best Next Experiment?

- Label panel: conservative-safe only.
- Batch size: 8 or 16.
- DAB weight: 0.02, 0.05, 0.1.
- GRL/DAB warm-up over first 30–50% of epochs.
- Early stopping by lower safe-batch probe plus preserved H/D probe.
- Then fine-tune H/D from the best corrected checkpoint and evaluate ExVal.

20 Short Answers To Likely Questions

Question	Short answer
Is this exact sequencing batch?	No. It is an externally enriched study/cohort batch-proxy annotation. Exact technical batch would require original run date, kit, center, instrument, and processing metadata.
Why use batch proxies then?	Because they are enough to test whether embeddings encode study/cohort signal, which is a major source of external validation shift.
Why not correct all high-confidence labels?	Because high-confidence study labels can still be disease-confounded. A real label can be unsafe.
What does batch probe prove?	It proves batch labels are predictable from frozen embeddings; it does not by itself prove causality or solve correction.
Why did broad adversarial correction increase batch probe?	Likely because labels are heterogeneous/confounded and the adversarial game can reorganize batch information rather than remove it.
Is batch correction finished?	No. We have a working diagnostic and proof-of-concept pipeline, but final correction needs schedule tuning and ExVal validation.
Why are 0.7–0.8 numbers appearing?	They are often AUROC or embedding-probe metrics, not ExVal macro-F1. The strict external macro-F1 is currently around 0.56–0.61 depending on threshold.
What is the most defensible current claim?	Taxonomy-aware BiomeGPT learns biologically meaningful sample embeddings, but external prediction is limited by calibration/domain shift; batch correction is motivated but must be conservative.